

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. Determine the domain and range of $f(g(x))$ where $f(x) =$

$\frac{1}{x+1}$ and $g(x) = x - 2$, and interpret the result in the context of a scoreboard system.

Q10. A gaming platform's user engagement E is determined by $E(x) = x^2 - 5x + 6$ where x is the number of active users. If x is restricted by $g(x) = x + 1$, find $E(g(x))$ and discuss its implications on the platform's design.

Chef's Tips

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- Tip 1: Ensure students understand the concept of composite functions before attempting the test.
- Tip 2: Use real-world examples to help students interpret the results of composite functions.
- Tip 3: Remind students to check the domain restrictions of composite functions to avoid undefined values.